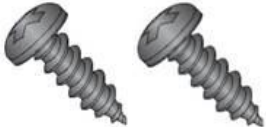


Switch Adapted Air Hogs Zero Gravity RC Car

MAKER GUIDE

Required Components

1. 	2. 	BOM 1. Air Hogs Zero Gravity Sprint RC Car 2. (4x) 3.5 mm Female Mono Cable 3. 3D Printed Cable Clamp 4. (2x) #4 3/8" Screws 5. (4x) Mini cable ties 6. (2x) AAA Batteries Refer to the Tool and Component List for purchasing details.
3. 	4. 	5. 
6. 		

Required Tools

- Small Phillips Screwdriver 3mm Diameter
- Wire strippers
- Soldering iron and solder
- Drill and 1/8" drill bit

Required Personal Protective Equipment (PPE)

- Safety glasses



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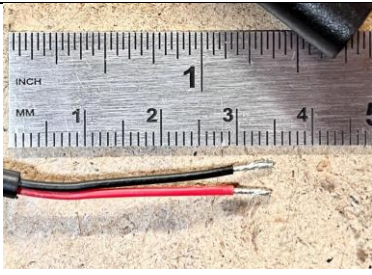
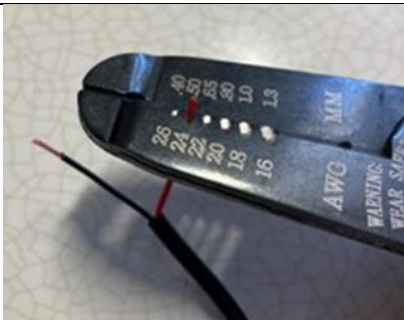
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Files available at https://github.com/makersmakingchange/Switch_Adapted_Toys

Switch Adapted Air Hogs Zero Gravity RC Car

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Assembly Instructions

1. Carefully remove the car and controller from its packaging without damaging it (we will be repackaging the toy after adapting). Locate and remove the screws along the body of the controller. <i>*Note: keep screws in a safe place</i>	
2. Carefully separate the two halves of the toy.	
3. To prepare the mono cable, use wire strippers. Strip enough outer wire sheathing to leave two 35 mm wires protruding. Repeat for all four mono cables.	
4. <u>If there are 2 internal wires</u>, strip approximately 3 mm or 1/8" off each wire.	

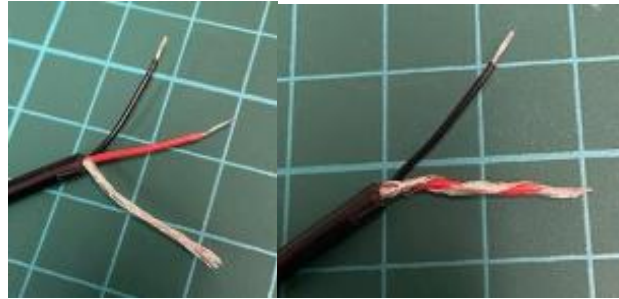
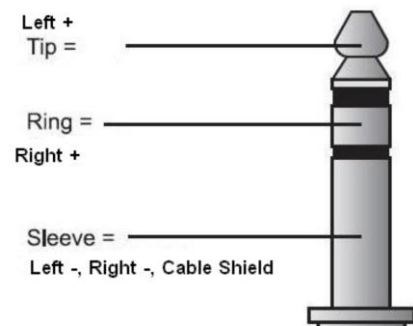
Switch Adapted Air Hogs Zero Gravity RC Car

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5. If there are 3 internal wires, you'll need to determine which wire is "Tip" "Ring" and "Sleeve". The most common wire colours are white (Tip), red (Ring) and green (Sleeve). If there is no White wire, then "Tip" may be black. If there is no green wire, the sleeve may be an UNCOATED silver or copper colour.

As cable wire colours can vary, have the wires tested with a multimeter if unsure.

Once you have determined which wire is which, strip off 3 mm from the end of each wire, and solder together the "Ring" and "Sleeve" wires, as shown here.



6. Tin the ends of the wires by adding a small amount of solder to each wire.



7. Using a 1/8" drill bit, drill 4 holes in the top half of the case into the positions shown on the right (See next panel for example). Remove any loose plastic.



Switch Adapted Air Hogs Zero Gravity RC Car

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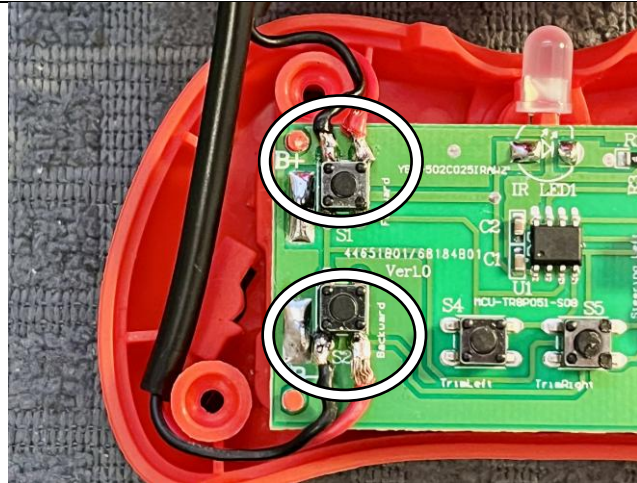
8. Insert the mono cables through the holes.



9. Solder the upper left case wires to the two contacts on the upper left switch, shown circled.

Solder the lower left case wires to the two contacts on the lower left switch, shown circled.

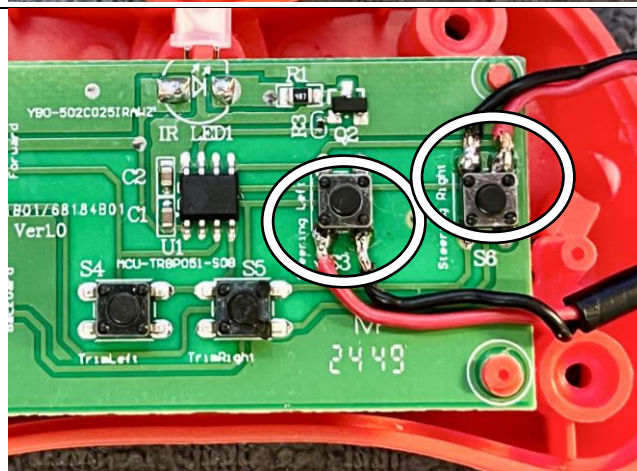
When you've soldered both left cables, the PCB should look like this:



10. Solder the upper right case wires to the two contacts on the rightmost switch, shown circled on the right.

Solder the lower right case wire to the two contacts on the adjacent switch to its left, shown in the left circle.

When you've soldered the two right-side cables, the PCB should look like this:



Switch Adapted Air Hogs Zero Gravity RC Car

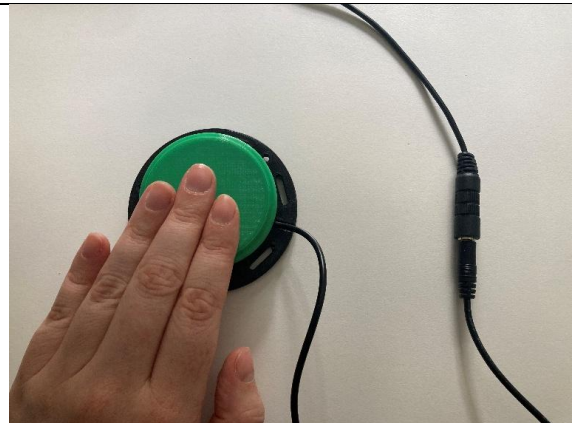
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11. Next, test the toy. Make sure both the remote and car have batteries inserted and that the car is turned on. Plug in an assistive switch. When you activate the switch, the car should perform the function of the button you soldered the cable to (Forward, Backward, Left, Right).

If the car does not move, make sure the batteries are pressed in fully and check the solder connections.

If the car does not stop moving when you release the switch, check the solder connections.

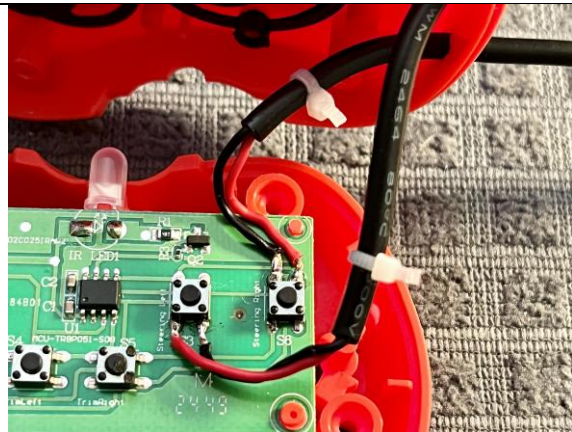
Repeat this for each cable.



12. Once the wires are connected, attach cable ties to the cable on the inside of the case as close as possible to where it enters.

Tighten the cable tie as much as you can and snip the excess cable tie. Leaving 3mm of strap on cable ties allows re-tightening with needle nose pliers.

The cable ties will serve as strain relief, so the soldered connections do not get pulled and break.



13. Begin reassembling the case. Pull the four cables out until they're stopped by the cable ties. Gently nestle down the lid while ensuring the wires don't get pinched under the black button presser panel, or any of the posts.



Switch Adapted Air Hogs Zero Gravity RC Car

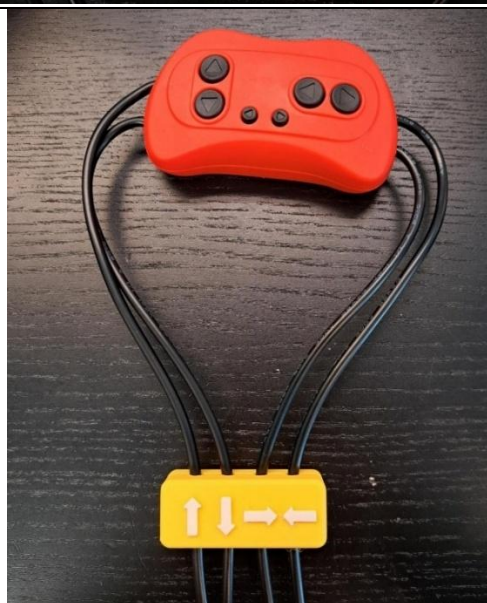
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14. Reassemble the toy using the screws set aside earlier.



15. Gather the mono cables, and put the cable clamp on, **ensuring that each wire is lined up with the proper symbol on the clamp.**

Insert the screws into the four holes on the back, and tighten to fasten the cable clamp.



16. Test each cable of the toy again. If it is not working properly, take apart the controller and ensure that none of the wires were pinched while assembling the controller. If it still doesn't work, check the soldered connections.

If it is still working properly, repackage the toy and the adapted toy is complete!

